

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2022-23

Zoology

Paper: ZOB-301 — Basic Genetics, Evolution and Developmental Biology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Basic Genetics and Evolution) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False, giving reasons for your answer:

$[2 \times 3 = 6]$

- (i) A $t(9;22)$ chromosome translocation is seen in chronic myeloid leukemia.
- (ii) Polygeny and pleiotropy refer to one and the same phenomenon.
- (iii) The idea of survival of the fittest was first proposed by Lamarck.

- (b) Define the following terms:

$[1 \times 2 = 2]$

- (i) Expressivity
- (ii) Atavism

- (c) Select the correct option in each of the following:

$[1.5 \times 2 = 3]$

- (i) Which one of the following is not the specific feature of an X-linked recessive pedigree?
 - (1) Disease affects mainly males.
 - (2) Affected males have unaffected parents.
 - (3) Affected males have affected fathers.
 - (4) Affected males can transmit the disease to their sons.
- (ii) Industrial melanism is an example of:
 - (1) Directional selection
 - (2) Disruptive selection
 - (3) Stabilizing selection
 - (4) Artificial selection

Question 2. Answer the following parts:

- (a) Discuss in detail the mechanism of sex determination in *Drosophila* and humans. [6]
- (b) Using suitable examples, explain the features of codominance and incomplete dominance. [6]

Question 3. Answer the following parts:

- (a) What is Hardy-Weinberg equilibrium? Discuss the various factors that disturb the Hardy-Weinberg equilibrium. [6]
- (b) Briefly write about the key concepts of Darwinism and the evolutionary significance of variations. [6]

Question 4. Write short notes on any two of the following: . . . [6 × 2 = 12]

- (a) Origin of life
- (b) Genetic drift
- (c) Chromosomal theory of inheritance

SECTION – B
(Developmental Biology) (Marks: 35)

Question 1. Answer the following parts:

- (a) Match the items in Column-I with Column-II: [1 × 5 = 5]

Column-I	Column-II
(i) Acrosomal reaction	(1) Morphogenetic movement
(ii) Cortical reaction	(2) <i>Amphioxus</i>
(iii) Epiboly	(3) Calcium wave
(iv) Area opaca	(4) Hyaluronidase
(v) Microlecithal egg	(5) Chick embryo

- (b) Define the following terms in one/two sentences each: [1 × 3 = 3]

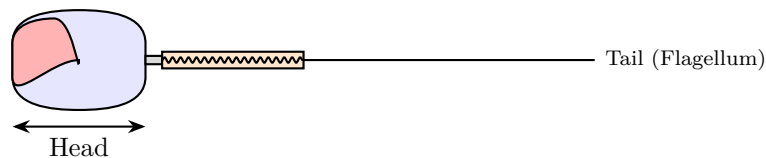
- (i) Epimorphosis
- (ii) Delamination
- (iii) Syncytial blastoderm

- (c) Fill in the blanks with appropriate words: [1 × 2 = 2]

- (i) The process by which cells become specialized in structure and function is called _____.
- (ii) Primitive streak is formed during the process of _____.

Question 2. Answer the following parts:

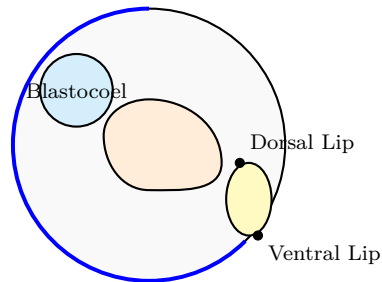
- (a) With the help of suitable diagrams, describe the process of spermiogenesis. [6]



- (b) What is cleavage? Describe the different types of cleavage on the basis of pattern with suitable examples. [6]

Question 3. Answer the following parts:

- (a) Give a detailed account of the process of gastrulation in frog with neat and labeled diagrams. [6]



- (b) Describe the different morphogenetic movements during gastrulation. [6]

Question 4. Write short notes on any two of the following:

[6 × 2 = 12]

- (a) Blastula of chick
- (b) Fast block to polyspermy
- (c) Regeneration in *Hydra*

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